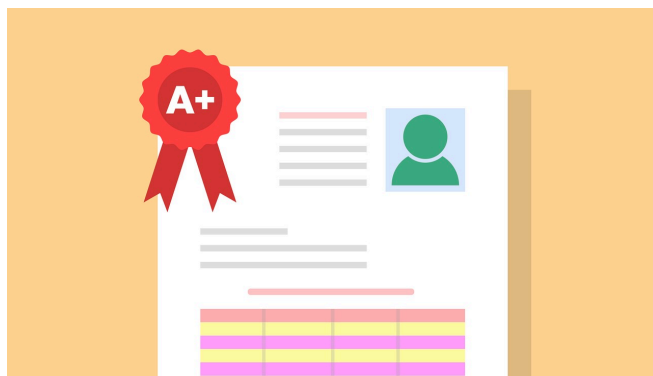




In today's fast-paced and competitive educational environment, understanding the factors that influence student success is more important than ever. Just like the transport system in a bustling city like London must adapt to serve its residents, schools and educators must adapt to meet the needs of students. In this project, we will take a deep dive into a dataset containing rich details about various aspects of student life, such as hours studied, sleep patterns, attendance, and more, to uncover what truly impacts exam performance.

The dataset we'll be working with includes a wide range of factors influencing student performance. By analyzing this data, we'll be able to identify key drivers of success and provide insights that could help students, teachers, and policymakers make informed decisions. The table we'll use for this project is called `student_performance` and includes the following data:

Column	Definition	Data type
<code>attendance</code>	Percentage of classes attended	<code>float</code>
<code>extracurricular_activities</code>	Participation in extracurricular activities	<code>varchar</code> (Yes, No)
<code>sleep_hours</code>	Average number of hours of sleep per night	<code>float</code>
<code>tutoring_sessions</code>	Number of tutoring sessions attended per month	<code>integer</code>
<code>teacher_quality</code>	Quality of the teachers	<code>varchar</code> (Low, Medium, High)
<code>exam_score</code>	Final exam score	<code>float</code>

You will execute SQL queries to answer three questions, as listed in the instructions.

 Projects Data DataFrame as `avg_exam_score_by_study_and_exti`

```
-- avg_exam_score_by_study_and_extracurricular
-- Edit the query below as needed
```

```
SELECT *
FROM student_performance
LIMIT 30;
```

i. ...	↑↓	hours_studied ...	↑↓	atten... ...	↑↓	extracurricular_activities ...	↑↓	sleep_h... ...	↑↓	tutoring_sessions ...
0		23		84		No		7		
1		19		64		No		8		
2		24		98		Yes		7		
3		29		89		Yes		8		
4		19		92		Yes		6		
5		19		88		Yes		8		
6		29		84		Yes		7		
7		25		78		Yes		6		
8		17		94		No		6		
9		23		98		Yes		8		
10		17		80		No		8		
11		17		97		Yes		6		
12		21		83		Yes		8		
13		9		82		Yes		8		
14		10		78		Yes		8		
15		17		68		No		8		

Rows: 30

 Expand

Projects Data Query as avg_exam_score_by_study_and_exti

```
select hours_studied, avg(exam_score) as avg_exam_score
from student_performance
group by hours_studied
having hours_studied > 10
order by hours_studied desc
limit 30;
```

index	...	↑↓	hours_studied	...	↑↓	avg_exam_score
		0			44	
		1			43	
		2			39	
		3			38	
		4			37	
		5			36	
		6			35	
		7			34	
		8			33	
		9			32	
		10			31	
		11			30	
		12			29	
		13			28	
		14			27	
		15			26	

Rows: 30

Expand

Projects Data Query as query

```
select
  case when hours_studied between 1 and 5 then '1-5 hours'
  when hours_studied between 6 and 10 then '6-10 hours'
  when hours_studied between 11 and 15 then '11-15 hours'
  else '16+' end as hours_studied_range,
  avg(exam_score) as avg_exam_score
from student_performance
group by hours_studied_range
order by avg(exam_score);
```

...	↑↓	hours_studied_ra...	...	↑↓	avg_exa...	...	↑↓
0		1-5 hours			62.6271186441		
1		6-10 hours			64.2254901961		
2		11-15 hours			65.2043859649		
3		16+			67.9233633869		

Rows: 4

Expand

Projects Data Query as s

```
select attendance, hours_studied, sleep_hours, tutoring_sessions,  
       rank() over(order by exam_score) as exam_rank  
from student_performance  
order by exam_rank  
limit 30;
```

...	↑↓	a...	...	↑↓	hours_...	...	↑↓	sle...	...	↑↓	tutoring_sessi...	...	↑↓	e.	...	↑↓
0					62			3			6			1		1
1					65			5			7			0		2
2					66			7			8			0		3
3					64			9			10			0		3
4					67			14			7			0		3
5					60			3			6			1		3
6					69			11			8			0		7
7					61			12			7			2		7
8					61			13			7			1		7
9					64			13			6			1		7
10					61			4			7			0		7
11					60			15			7			0		7
12					60			3			5			0		7
13					63			11			9			0		7
14					61			9			7			0		7
15					62			4			6			3		7

Rows: 30

[↗ Expand](#)